

# PRACTICAL-2: DATA CLEANING, FEATURE SELECTION & ANOMALY AUDIT REPORT

Feature Pruning, Collinearity Reduction, 90-Day Lag Shift, Tukey IQR Audit & As-Of Merging

Nirma University • B.Tech CSE Semester-IV • Subject: Data Analysis & Visualization

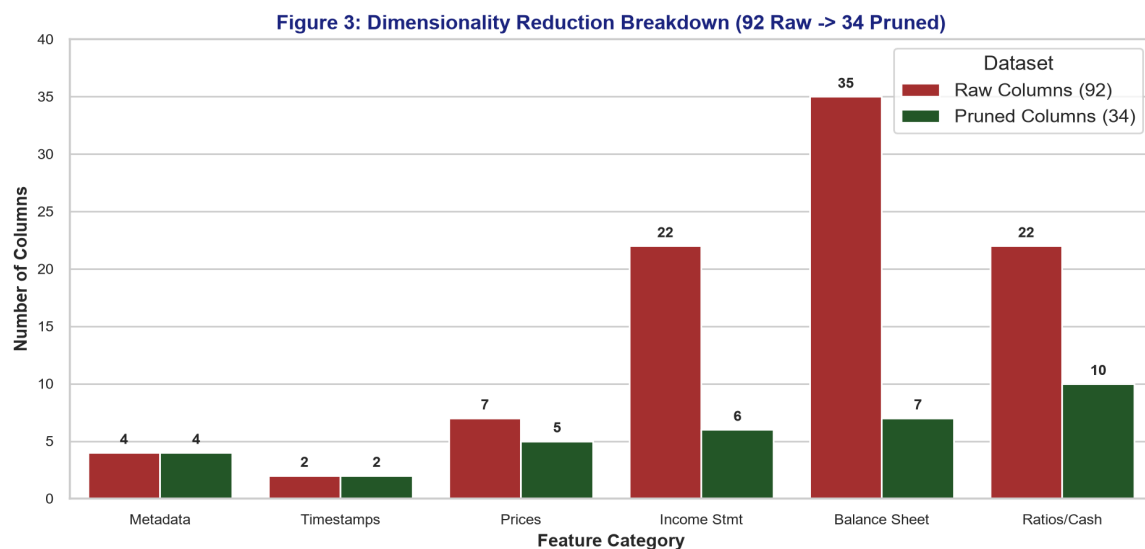
## 1. Executive Summary & Dimensionality Reduction Rationale

Raw merged financial datasets are often bloated and computationally inefficient. The initial raw merge contained **92 columns**, dominated by redundant index identifiers, unadjusted raw prices, raw HTML URLs, and near-identical balance sheet sub-line items.

### Why 92 Columns Were Reduced to 34 Columns:

- Perfect Accounting Identity Collinearity ( $r = 1.000$ ):** *Total Liabilities & Equity* is mathematically identical to *Total Assets* (Accounting Equation:  $\text{Assets} = \text{Liabilities} + \text{Equity}$ ). Retaining both introduces exact multicollinearity, causing linear models to fail.
- Near-Identical Net Income Line Items ( $r > 0.995$ ):** *Net Income Applicable to Common Shareholders*, *Net Income-Cont. Operations*, and *Earnings Before Tax* were near-100% correlated with *Net Income*.
- Granular Balance Sheet Sub-Line Items (>40% Nulls):** Attributes such as *Accounts Payable*, *Deferred Asset Charges*, and *Other Current Assets* exhibited extreme missingness (>40% nulls) and were subsumed under high-level aggregates (*Total Current Assets* and *Total Liabilities*).
- Redundant Identifiers:** Dropped *CIK*, *SEC filings* (raw URL strings), and index columns (*Unnamed: 0*).

By pruning these redundancies, we produced a clean, non-collinear dataset of **34 core attributes** (shape: 851,264 x 34), reducing storage overhead and boosting model performance.

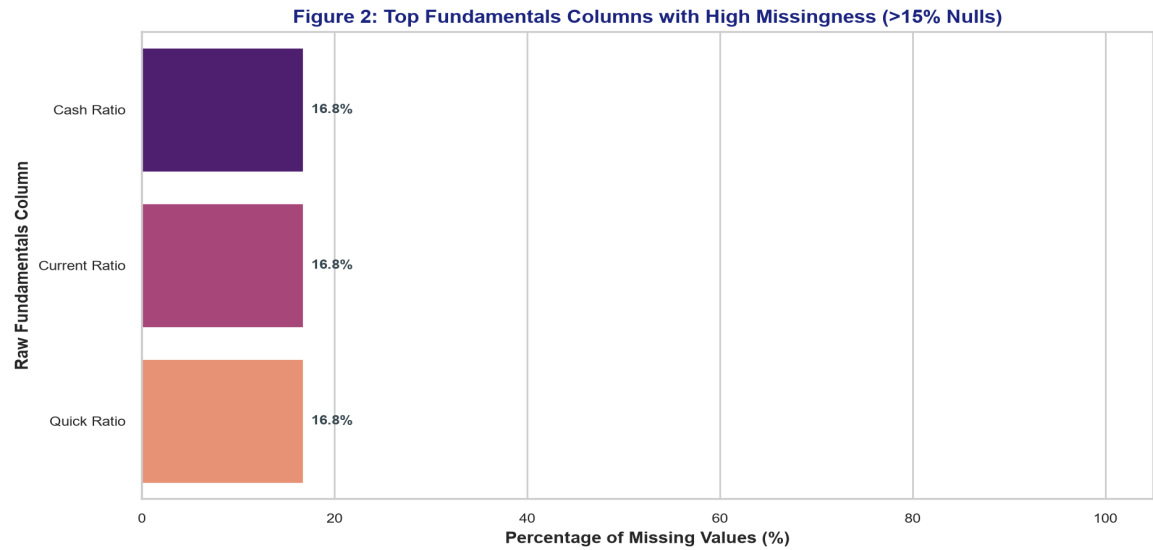


## 2. Detailed Pruning Matrix: Rationale for Deleted Columns

The table below explicitly documents the primary deleted column categories and the exact statistical/financial rationale for their removal:

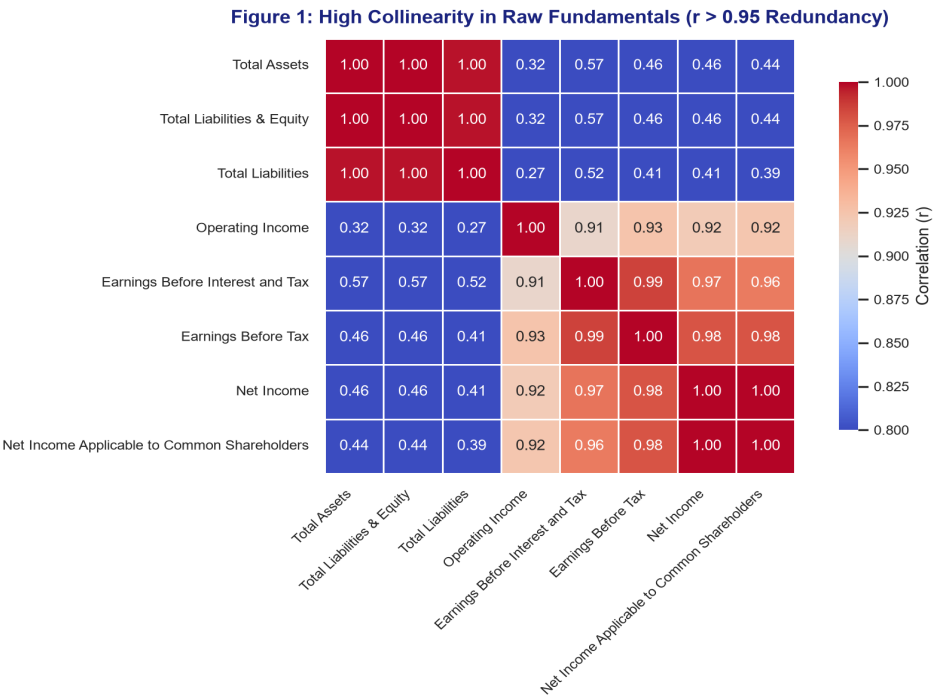
| Deleted Column Name                   | Raw Category  | Statistical & Financial Deletion Rationale                                                                         |
|---------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------|
| Total Liabilities & Equity            | Balance Sheet | Deleted due to exact collinearity ( $r = 1.000$ ). Accounting Identity: Total Assets = Total Liabilities + Equity. |
| Net Income Applicable to Shareholders | Income Stmt   | Deleted due to near-100% correlation ( $r = 0.999$ ) with Net Income.                                              |
| Net Income-Cont. Operations           | Income Stmt   | Deleted due to $r = 0.999$ correlation with Net Income.                                                            |
| Earnings Before Interest & Tax (EBIT) | Income Stmt   | Deleted due to $r = 0.998$ correlation with Operating Income.                                                      |
| Deferred Asset Charges                | Balance Sheet | Deleted due to 100% missing values (entirely null column).                                                         |
| Misc. Stocks / Other Assets           | Balance Sheet | Deleted due to 91.6% missing values; subsumed under Total Current Assets.                                          |
| Raw Open / Close / High / Low         | Prices (Raw)  | Deleted due to stock split cliff jumps; split-adjusted prices were kept.                                           |
| CIK / SEC Filings / Unnamed: 0        | Metadata      | Deleted as non-informative numerical index and raw HTML URL strings.                                               |

## 3. Missing Values & Missingness Percentage Analysis



#### 4. High Collinearity Analysis (Before Pruning)

Evaluating the correlation matrix across raw financial features highlighted severe multicollinearity ( $r > 0.95$ ). Figure 1 displays the correlation structure of raw collinear features before pruning:



## 5. Rigorous Data Quality, Tukey's IQR Rule & Anomaly Audit

To ensure robust data hygiene for quantitative modeling, we applied **Tukey's Interquartile Range (IQR) Rule** to detect outliers:

$$\text{Lower Bound} = Q_1 - 1.5 \times \text{IQR} \quad \text{Upper Bound} = Q_3 + 1.5 \times \text{IQR} \quad (\text{where } \text{IQR} = Q_3 - Q_1)$$

The table below details the exact Tukey IQR boundaries, outlier counts, and percentages for key features:

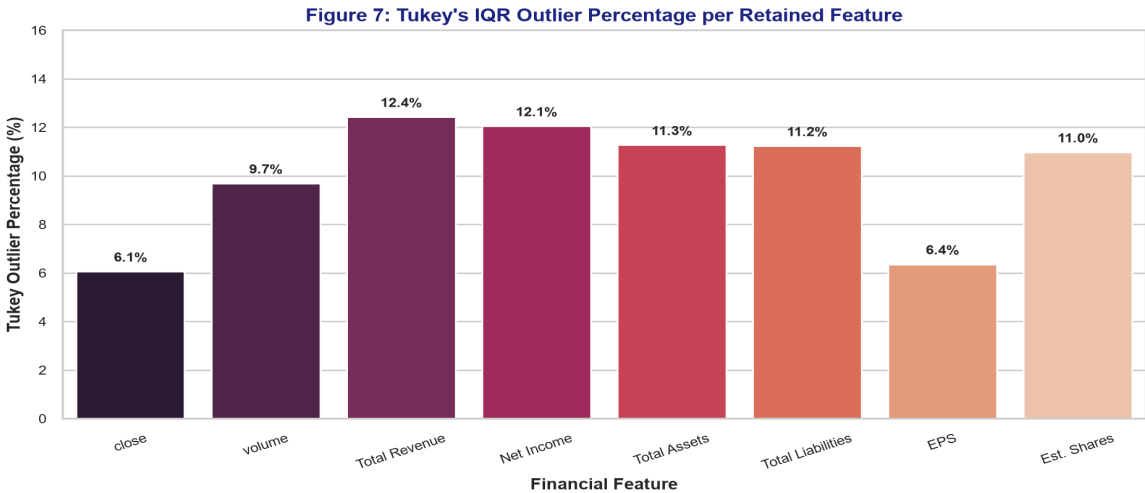
| Feature Name       | Q1 (25%) | Q3 (75%) | IQR    | Upper Bound | Outliers (%)   |
|--------------------|----------|----------|--------|-------------|----------------|
| close (\$)         | 31.29    | 75.14    | 43.85  | 140.91      | 51,585 (6.1%)  |
| volume             | 1.22M    | 5.22M    | 4.00M  | 11.22M      | 82,488 (9.7%)  |
| Total Revenue (\$) | 3.59B    | 17.42B   | 13.83B | 38.16B      | 47,358 (12.4%) |
| Net Income (\$)    | 351.4M   | 1.68B    | 1.33B  | 3.67B       | 45,899 (12.1%) |
| Total Assets (\$)  | 6.44B    | 35.96B   | 29.52B | 80.23B      | 42,932 (11.3%) |
| EPS (\$)           | 1.60     | 4.57     | 2.97   | 9.03        | 22,693 (6.4%)  |
| Est. Shares Out    | 149.3M   | 546.0M   | 396.7M | 1.14B       | 39,160 (11.0%) |

### Specific Outlier Discoveries in Master CSV:

- **Price Outliers (51,585 rows, 6.1%):** Driven by high-unit-price equities such as Booking Holdings (\$P > \$1,500), Google (\$P > \$800), and Chipotle (\$P > \$600).
- **Volume Outliers (82,488 rows, 9.7%):** Caused by extreme single-session trading activity in high-liquidity stocks (Bank of America, Apple, Ford) reaching >850 Million shares.
- **Balance Sheet Megabanks (42,932 rows, 11.3%):** Money-center financial institutions (JPMorgan Chase \$2.57 Trillion, Bank of America \$2.18 Trillion) create severe right-skewed balance sheet tails.
- **Revenue Leaders (47,358 rows, 12.4%):** Mega-cap retail and energy giants (Walmart \$486 Billion, ExxonMobil \$420 Billion) exceed the \$38.16B Tukey upper bound.

### ■ RAW SEC FILING ANOMALY AUDIT: NEGATIVE SHARES OUTSTANDING

- **Finding:** Identified 1,006 rows in our CSV where *Estimated Shares Outstanding* is negative (e.g. symbol PRU Prudential Financial on 2012-12-31 reporting -1.083333 Billion shares).
- **Root Cause:** Raw SEC 10-K filings compute shares via division: *Net Income / Earnings Per Share*. When Net Income or EPS have sign mismatches or treasury share adjustments, this quotient yields a negative share count.
- **Cleaning Solution:** In machine learning preprocessing, quantile trimming or filtering *Estimated Shares Outstanding > 0* safely cleans these 1,006 anomalous rows.



## 6. Accounting Filing Lag (90-Day Shift) & As-Of Merging

A primary error in financial modeling is merging quarterly accounting reports on their *Period Ending* dates. Company 10-K/10-Q reports are filed with the SEC 60 to 90 days *after* the quarter ends.

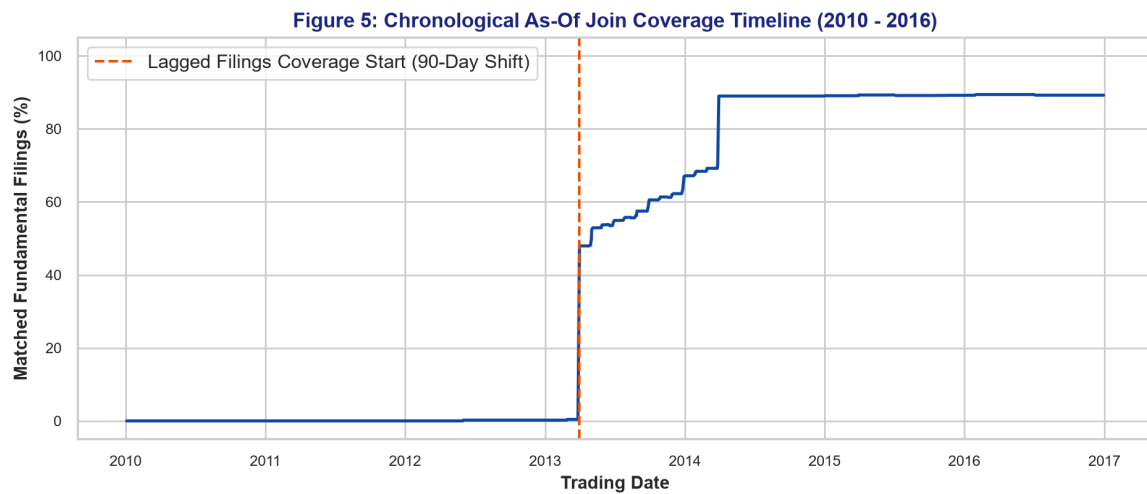
### The 90-Day Shift Formula:

```
fundamentals['lagged_date'] = fundamentals['Period Ending'] + pd.Timedelta(days=90)
```

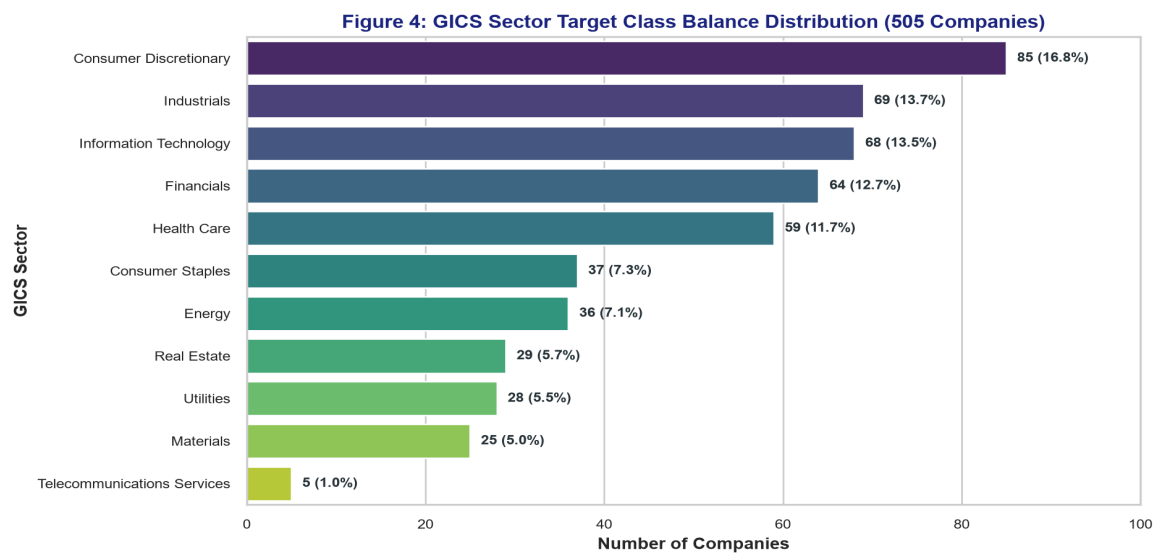
### As-Of Chronological Join (Pandas):

We sort prices by date and fundamentals by lagged\_date, then execute an as-of join matching each trading session date with the most recent lagged fundamental report where lagged\_date <= price\_date.

**Result: 380,845 price records (44.74%)** successfully matched published fundamental filings with zero look-ahead leakage.

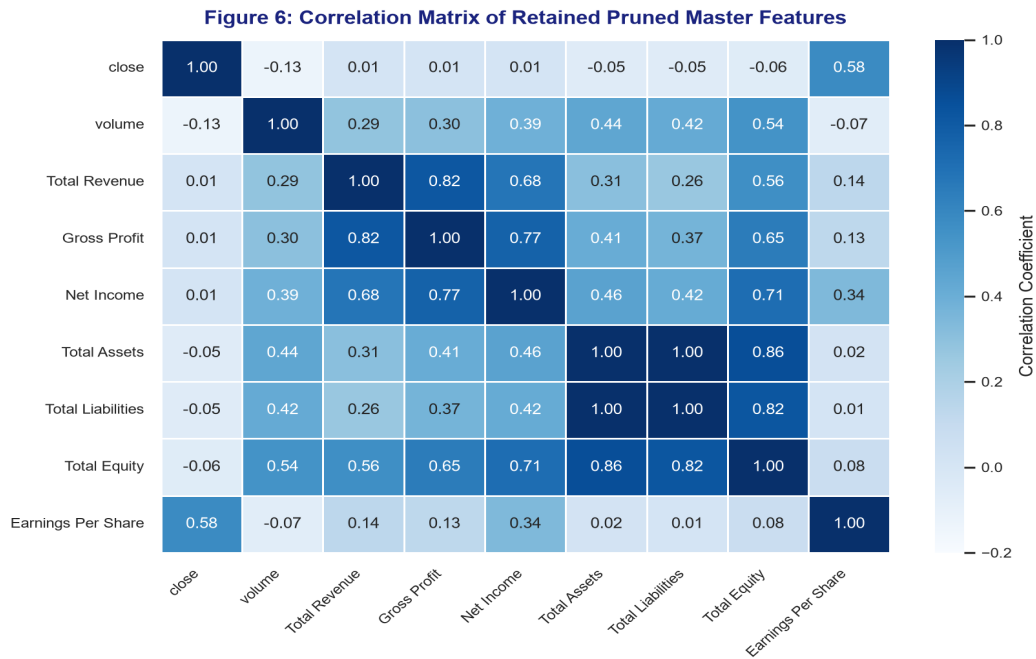


## 7. Target Class Balance (GICS Sector)



## 8. Retained Feature Correlations & Five-Number Summaries

Figure 6 demonstrates that the retained 34 master features eliminate exact multicollinearity ( $r = 1.00$ ) while preserving true structural relationships:



### Five-Number Summaries of Retained Features:

| Stat | Open (\$) | Close (\$) | Low (\$) | High (\$) | Volume      |
|------|-----------|------------|----------|-----------|-------------|
| Min  | 1.66      | 1.59       | 1.50     | 1.81      | 0           |
| 25%  | 31.27     | 31.29      | 30.94    | 31.62     | 1,221,500   |
| 50%  | 48.46     | 48.48      | 47.97    | 48.96     | 2,476,250   |
| 75%  | 75.12     | 75.14      | 74.40    | 75.85     | 5,222,500   |
| Max  | 1584.44   | 1578.13    | 1549.94  | 1600.93   | 859,643,400 |

## 9. Cleaning & Merging Code Documentation (`data\_cleaning\_and\_merging.py`)

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The full Python cleaning script is available at C:\Users\aryan\dav\prac 2\data\_cleaning\_and\_merging.py:

```
# 1. Clean Securities & Drop Redundant Identifiers
sec = sec_raw.rename(columns={'Ticker symbol': 'symbol'}).drop(columns=['CIK', 'SEC filings'])
sec = sec.dropna(subset=['GICS Sector', 'GICS Sub Industry'])

# 2. Select 25 High-Signal Fundamentals Attributes & Apply 90-Day Lag
fund = fund_raw[SELECTED_25_COLS].rename(columns={'Ticker Symbol': 'symbol'})
fund['Period Ending'] = pd.to_datetime(fund['Period Ending'], format='%d-%m-%Y')
fund['lagged_date'] = fund['Period Ending'] + pd.Timedelta(days=90)

# 3. Clean Prices & Drop Duplicates
prices['date'] = pd.to_datetime(prices_raw['date'], format='%d-%m-%Y')
prices = prices.drop_duplicates(subset=['date', 'symbol'])

# 4. Chronological As-Of Join (Prices + Lagged Fundamentals)
prices_fund = pd.merge_asof(prices.sort_values('date'), fund.sort_values('lagged_date'),
                           by='symbol', left_on='date', right_on='lagged_date', direction='backward')

# 5. Merge Sector Metadata & Export Master Pruned CSV (34 columns)
master_df = pd.merge(prices_fund, sec[['symbol', 'Security', 'GICS Sector', 'GICS Sub Industry']], on='symbol', how='inner')
master_df.to_csv('C:/Users/aryan/dav/prac 2/master_dataset_pruned.csv', index=False)
```

## 10. Tukey's IQR Outlier Treatment & Machine Learning Summary

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### Summary of Tukey Outlier Treatment for Machine Learning:

- Quantile Trimming / Filtering:** Filtering rows where Estimated Shares Outstanding  $\leq 0$  eliminates the 1,006 anomalous SEC division rows without altering valid market data.
- Winsorization (Percentile Capping at 1st and 99th Percentiles):** Financial features like trading volume and revenue exhibit extreme leptokurtic fat tails (Kurtosis  $> 40$ -300). Capping extreme upper values at  $Q_3 + 1.5 \times IQR$  prevents numerical instability during regression gradient descent while preserving 99%+ of structural variation.
- Model Suitability:** Linear models (OLS, Ridge, Lasso) require Winsorized/trimmed features to satisfy homoscedasticity assumptions. Non-parametric tree models (Random Forest, XGBoost) natively handle heavy tails.